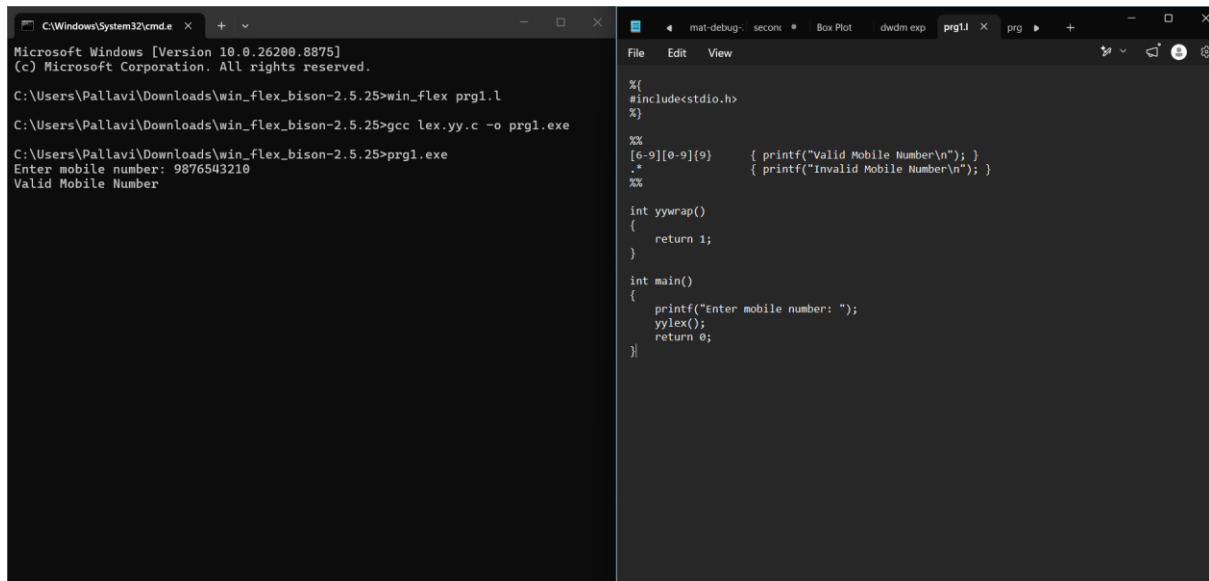


Lab programs(lexical):

Program 1:



The screenshot shows two windows. The left window is a Windows Command Prompt with the following text:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26280.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg1.l
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg1.exe
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg1.exe
Enter mobile number: 9876543210
Valid Mobile Number
```

The right window is a code editor showing the source code for prg1.l:

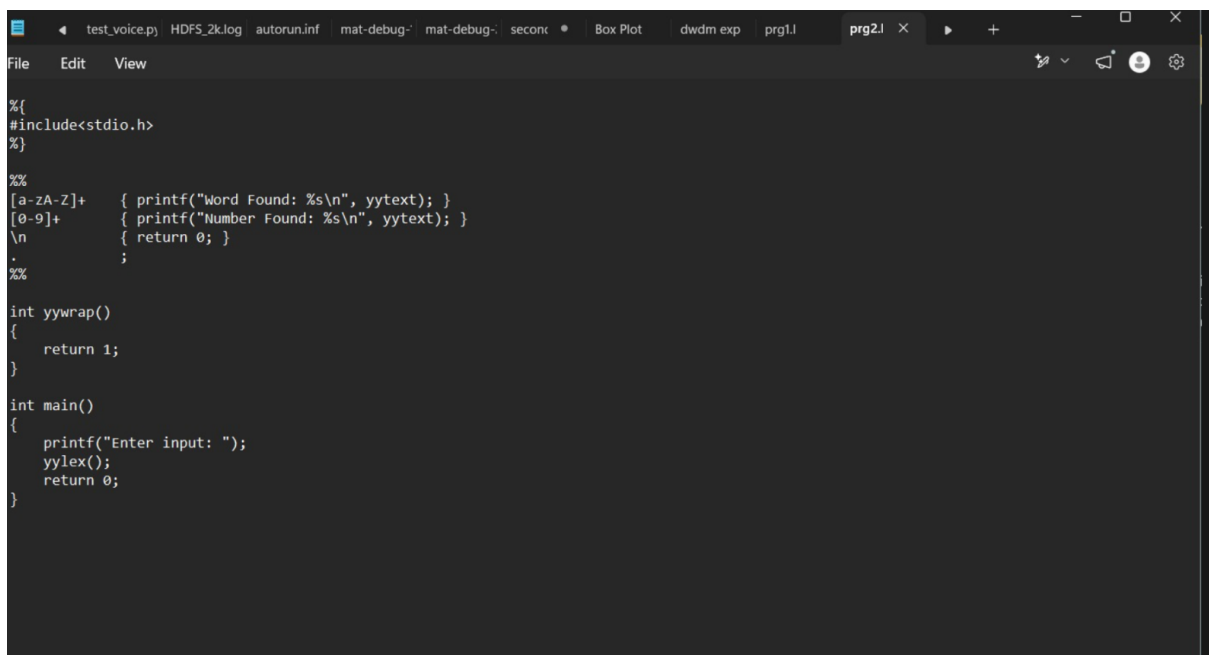
```
%{
#include<stdio.h>
}%

%%
[6-9][0-9]{9} { printf("Valid Mobile Number\n"); }
.* { printf("Invalid Mobile Number\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter mobile number: ");
    yylex();
    return 0;
}
```

Program 2:



The screenshot shows a code editor window with the following source code for prg2.l:

```
%{
#include<stdio.h>
}%

%%
[a-zA-Z]+ { printf("Word Found: %s\n", yytext); }
[0-9]+ { printf("Number Found: %s\n", yytext); }
\n { return 0; }
. ;
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter input: ");
    yylex();
    return 0;
}
```

```

Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg2.l

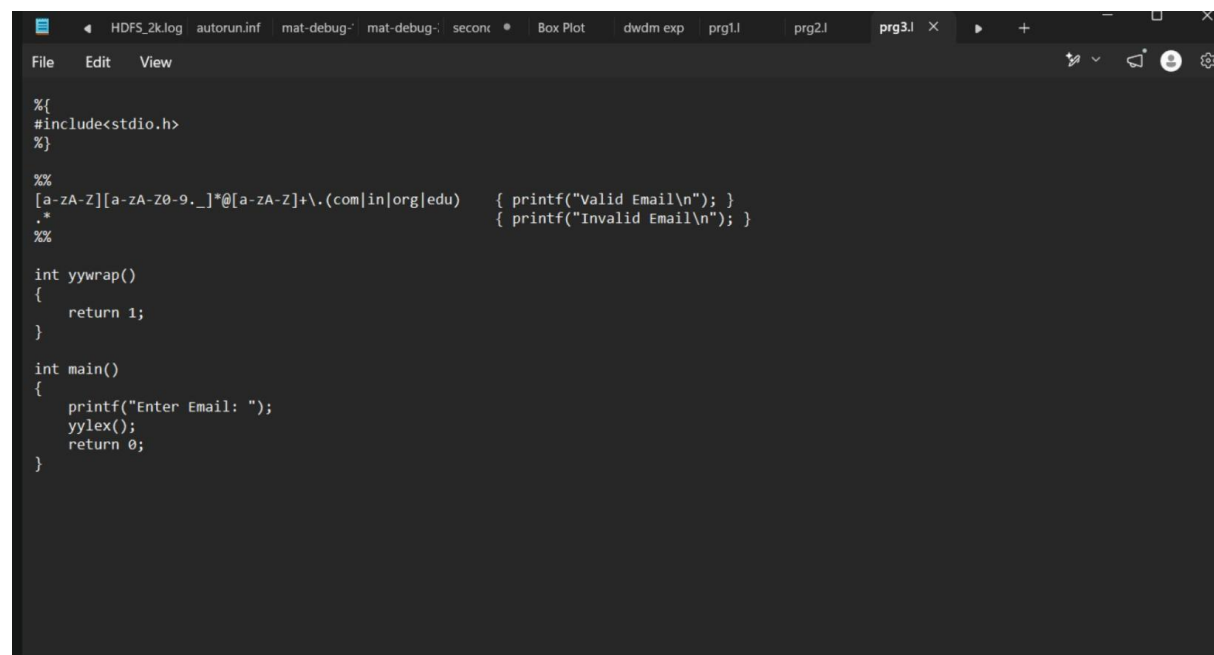
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg2.exe

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg2.exe
Enter input: hello 123
Word Found: hello
Number Found: 123

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>

```

Program 3:



```

File Edit View
%{
#include<stdio.h>
%}

%%
[a-zA-Z][a-zA-Z0-9._]*@[a-zA-Z]+\.(com|in|org|edu) { printf("Valid Email\n"); }
.* { printf("Invalid Email\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter Email: ");
    yylex();
    return 0;
}

```

```

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg3.l

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg3.exe

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg3.exe
Enter Email: abc@gmail.com
Valid Email

```

Program 4:

```
C:\Windows\System32\cmd.e x + v

prg4.l:7: unrecognized rule
prg4.l:7: unrecognized rule
prg4.l:7: unrecognized rule

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg4.l
prg4.l:7: warning, rule cannot be matched

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg4.exe

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg4.exe
Enter URL: http://google.com
Valid URL
|

File Edit View
%{
#include<stdio.h>
}%

%%
http.* { printf("Valid URL\n"); }
https.* { printf("Valid URL\n"); }
.* { printf("Invalid URL\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter URL: ");
    yylex();
    return 0;
}

Ln 21, Col 2 | 251 characters | Plain text | 100% | Windows (CRLF) | UTF-8

prg4
```

Program 5:

The screenshot shows a Windows command prompt and a code editor. The command prompt shows the compilation of `lex.yy.c` to `prg5.exe` and its execution. The program prompts the user to enter a string, which is "Computer Science". It then counts the vowels (o, u, e, i, e, e) and prints "Total Vowels = 6". The code editor shows the source code for `prg5.l`, which uses flex to tokenize the input string and count the vowels.

```
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg5.exe
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg5.exe
Enter a string: Computer Science
Vowel: o
Vowel: u
Vowel: e
Vowel: i
Vowel: e
Vowel: e
Total Vowels = 6
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>|
```

```
%{
#include<stdio.h>
int count = 0;
}%

%%
[aAeEiIoOuU] { printf("Vowel: %s\n", yytext);
               count++;
               }
\n { printf("Total Vowels = %d\n", count);
    return 0;
    }
.
;

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter a string: ");
    yylex();
    return 0;
}
```

Program 6:

The screenshot shows a Windows command prompt and a code editor. The command prompt shows the compilation of `lex.yy.c` to `prg6.exe` and its execution. The program prompts the user to enter a DOB (DD/MM/YYYY), which is "05/11/2007". It then validates the DOB and prints "Valid DOB". The code editor shows the source code for `prg6.l`, which uses flex to tokenize the input string and validate the DOB.

```
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg6.l
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg6.exe
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg6.exe
Enter DOB (DD/MM/YYYY): 05/11/2007
Valid DOB
|
```

```
%{
#include<stdio.h>
}%

%%
[0-9][0-9]\/[0-9][0-9]\/[0-9][0-9][0-9][0-9] { printf("Valid DOB\n"); }
.+ { printf("Invalid DOB\n"); }
}%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter DOB (DD/MM/YYYY): ");
    yylex();
    return 0;
}
```

Program 7:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg6.l
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg6.exe
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg6.exe
Enter DOB (DD/MM/YYYY): 05/11/2007
Valid DOB

C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>win_flex prg7.l
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>gcc lex.yy.c -o prg7.exe
C:\Users\Pallavi\Downloads\win_flex_bison-2.5.25>prg7.exe
Enter Input: HELLO
Capital Letters
student
Valid Identifier
```

```
File Edit View
%{
#include<stdio.h>
}%

%%
[A-Z]+ { printf("Capital letters\n"); }
[a-zA-Z_][a-zA-Z0-9_]* { printf("Valid Identifier\n"); }
.* { printf("Invalid Identifier\n"); }
%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter Input: ");
    yylex();
    return 0;
}
```

exp7